

Weekend of Code 9.0

Turing's Playground Abstract

TEAM: *FAADU SYSTEM*

TEAM MEMBERS:

Member 1: *Iman Sardar, 2025CA045, CSED* *imansardar8000-source (Team Leader)*

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PROBLEM	STATEMENT	ID:
<i>A01</i>		

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Objective:

To build an AI-powered system that helps users detect and avoid online scams (UPI frauds, phishing messages, fake job offers, and scam links) in real time, before they lose money or personal data.

DESCRIPTION:

AI ScamShield is a web-based fraud detection platform that allows users to paste suspicious messages, links, or upload screenshots and instantly receive a scam risk score with clear explanations. The system uses machine learning models for natural language understanding and image analysis to identify scam patterns, fake domains, and manipulated payment screenshots, making digital safety accessible to everyone.

- **OUTLINE OF THE PROJECT :**
DATA INPUT

- *USER INPUTS TEXT (SMS, WHATSAPP, EMAIL)*
- *USER PASTES SUSPICIOUS URLS*
- *USER UPLOADS IMAGES (UPI SCREENSHOTS, FAKE OFFERS)*

- ***PREPROCESSING***

- *CLEAN AND NORMALIZE TEXT*
- *EXTRACT URLS AND KEY ENTITIES (AMOUNTS, BANKS, URGENCY WORDS)*
- *RESIZE AND NORMALIZE IMAGES FOR ANALYSIS*

- ***ML MODELS USED***

- *NLP MODEL TO CLASSIFY SCAM VS SAFE MESSAGES*
- *VISION MODEL TO DETECT FAKE OR MANIPULATED SCREENSHOTS*
- *AI-BASED LINK ANALYZER TO DETECT PHISHING DOMAINS*

- ***INFERENCE & SCORING***

- *EACH INPUT IS GIVEN A SCAM PROBABILITY (0–100%)*
- *REASONS FOR THE DECISION ARE GENERATED*

- ***FRONTEND***

- *USER-FRIENDLY WEB INTERFACE*

- *UPLOAD, PASTE, AND VIEW RESULTS INSTANTLY*
- **BACKEND**
 - *API BUILT USING FASTAPI TO CONNECT FRONTEND AND AI MODELS*

LIBRARIES USED:

FastAPI – backend API

OpenAI API – NLP and image analysis

Python – core backend

OpenCV / Pillow – image preprocessing

Requests – API calls

React / HTML-CSS-JS – frontend

Chart.js (optional) – visualizing risk score

HIGHLIGHTS:

- ***Detects UPI fraud, phishing, fake job offers, and scam messages***
- ***Uses real machine learning models (NLP + Vision)***
- ***Gives risk score + human-readable explanation***
- ***Works with text, links, and images***
- ***Designed specifically for Indian digital scams***
- ***Fully real-time and interactive***